Supporting information for

Structural Determination of Catalytically Active Subnanometer Iron Oxide Clusters

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Table S1. Comparison of catalytic performance between the iron-ceria (15R) and other Fe-based catalysts reported.

Sample	T / °C P / MPa	FTY·10^{-5} $mol_{CO}\ g_{Fe}^{-1}\ s^{-1}$	CO_2	Selectivity / wt %			Reference
				CH_4	C_2-C_4	C_{5+}	
15R	**250, 2.0**	**1.4**	**23**	**19**	**41**	**40**	**In this work**
Fe/NCNTs	300, 0.1	2.6	18.6	22.2	52.4	25.4	1
20Fe-100Al	260, 2.0	1.3	21.8	22.9	40.9	36.2	2
FeSi	270, 1.5	-	25.8	20	42.5	37.5	3
FeSi	260, 1.5	-	3.5	17.4	34.0	48.6	4

Table S2. Catalytic reactivity 15R sample for FTS with a CO conversion of 10.3 %..

Sample	T (°C)	CO Conv. (%)	CO_2	Selectivity (wt. %)[a]			
				CH_4	C_2-C_4	C_{5+}	Olefin[b]
15R[c]	230	10.3	14	20	27	53	68

[a] Based on carbon calculations for all hydrocarbons (exclude CO_2) [b] Olefin selectivity is calculated by $C_2^{=}$-$C_4^{=}$/C_2-C_4. [c] Reaction condition 0.2 g catalyst, H_2/CO = 2/1, 2.0 MPa, 6 mL·min^{-1}, time on stream of 5 h.

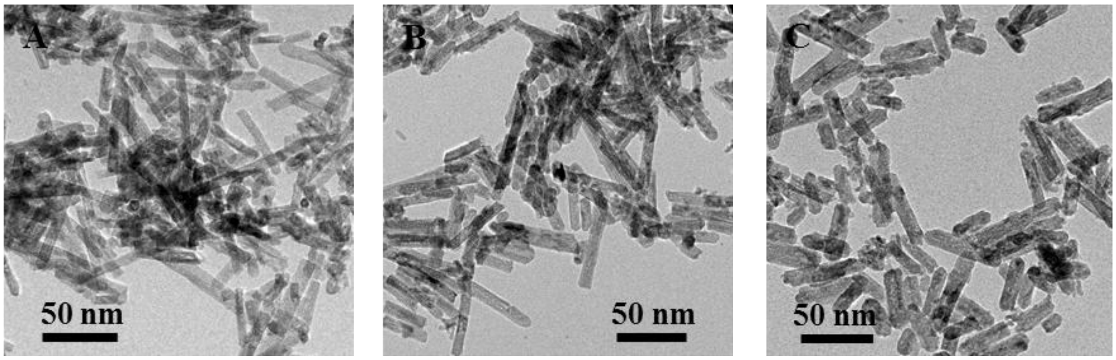

Figure S1. TEM images of the fresh NR samples: (A) **5R**, Fe/Ce=5/95; (B) **15R**, Fe/Ce=15/85; (C) **25R**, Fe/Ce=25/75.

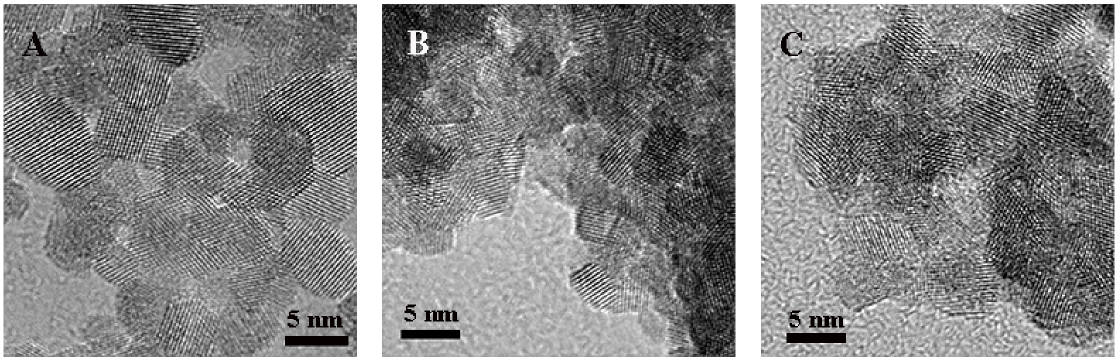

Figure S2. TEM images of the fresh NP samples: (A) **5P**, Fe/Ce=5/95; (B) **15P**, Fe/Ce=15/85; (C) **25P**, Fe/Ce=25/75.

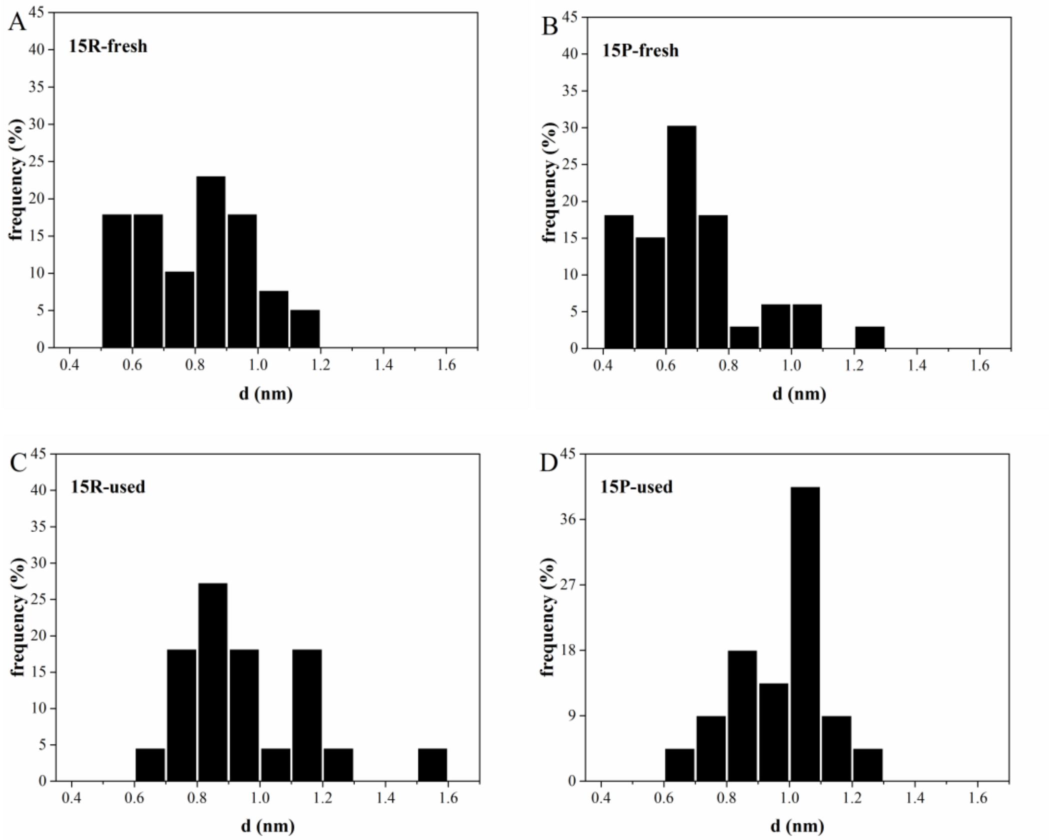

Figure S3. The histograms on cluster size of iron oxide in STEM-EELS: (A) **15R**, fresh; (B) **15P**, fresh; (C) **15R**, used; (D) **15P**, used.

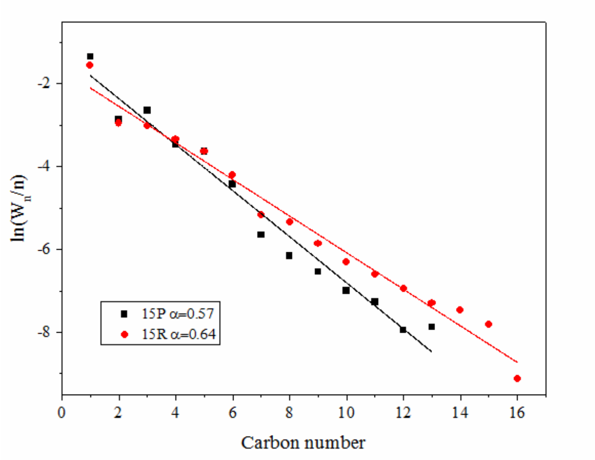

Figure S4. Anderson–Schulz–Flory plots of **15R** and **15P**: Iron-based catalysts were tested at 230 ^{o}C, H_2/CO = 2/1, 2.0 MPa, 6 mL·min^{-1}, time on stream of 5 h.

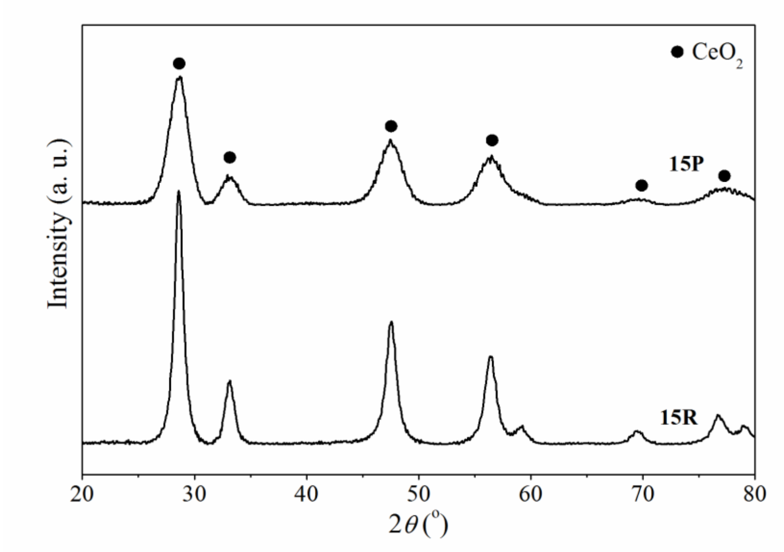

Figure S5. XRD patterns of used iron-ceria samples of **15P** and **15R.**

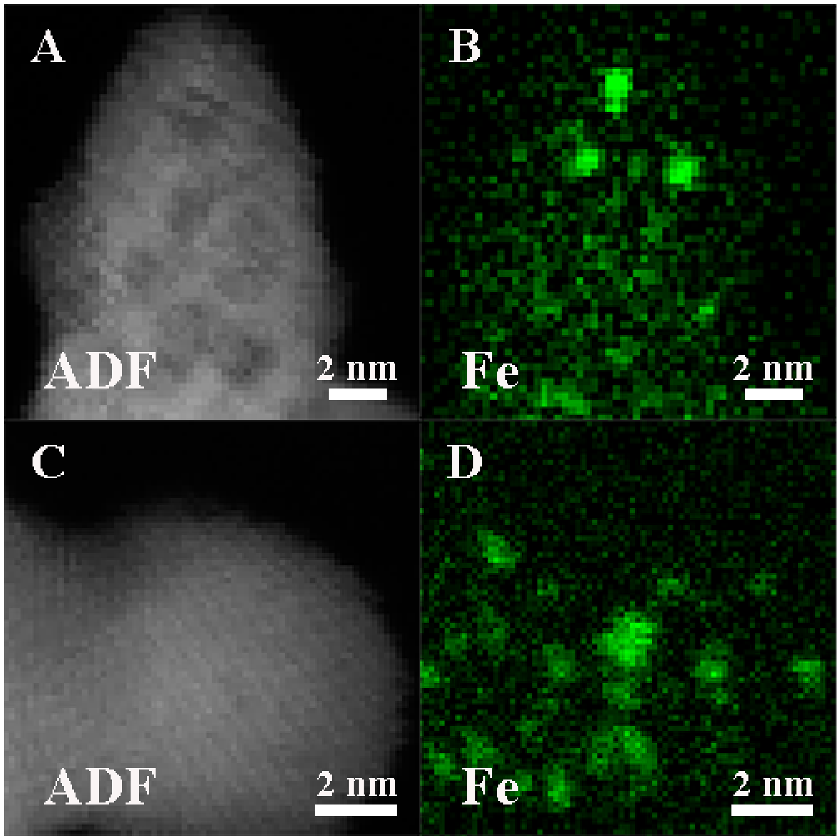

Figure S6. Aberration-corrected STEM-EELS results of iron-ceria samples after the H_2 pretreatment: (A, B) **15R**; (C, D) **15P**.

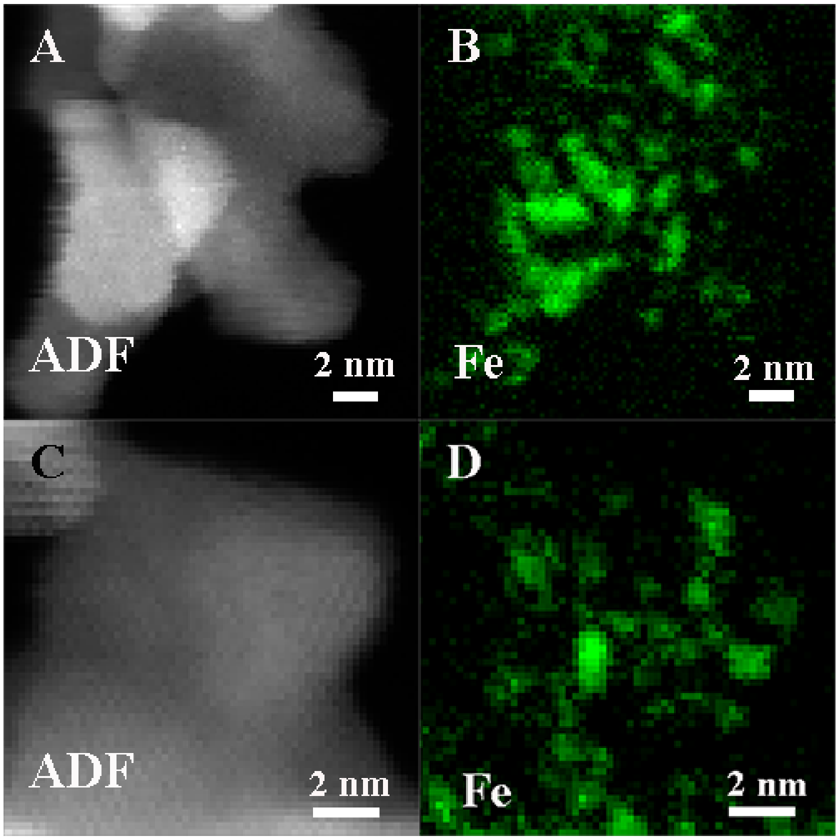

Figure S7. Aberration-corrected STEM-EELS results of used iron-ceria samples after the long-term tests with reaction time of 48 h: (A, B) **15R**; (C, D) **15P**.

Reference:

(1) Lu, J.; Yang, L.; Xu, B.; Wu, Q.; Zhang, D.; Yuan, S.; Zhai, Y.; Wang, X.; Fan, Y.; Hu, Z. *ACS Catal* **2014**, *4*, 613-621.

(2) Dong, H.; Xie, M.; Xu, J.; Li, M.; Peng, L.; Guo, X.; Ding, W. *Chem Commun* **2011**, *47*, 4019-21.

(3) Qing, M.; Yang, Y.; Wu, B.; Xu, J.; Zhang, C.; Gao, P.; Li, Y. *J. Catal* **2011**, *279*, 111-122.

(4) Li, J.; Zhang, C.; Cheng, X.; Qing, M.; Xu, J.; Wu, B.; Yang, Y.; Li, Y. *Appl. Catal. A* **2013**, *464-465*, 10-19.